Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

CTRL CV

Selected Domain

Domain 1 - AIVERSITY

Selected SDG Track (1–17)

Quality Education

Project / Solution Name

AI Workforce Pathon

Name (Leader)

MUHAMMAD ASYNAWI BIN AZWARRUDIN

Email & Phone (Leader)

asynawiazwarrudin@gmail.com

University Affiliation(s)

Universiti Teknikal Malaysia Melaka

Number of Team Members

4

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In Malaysia, rapid artificial intelligence adoption is outpacing traditional training frameworks. While 56% of Malaysian professionals already utilize AI tools at work, only 26% have received formal institutional or employer-led training (Hays Asia Salary Guide 2025). The Ministry of Science, Technology and Innovation (MOSTI) estimates that over 600,000 workers will require substantial reskilling within three to five years to avoid structural displacement.

This vulnerability affects both working adults facing automation and fresh TVET/university graduates entering an unpredictable labor market across tech and non-tech sectors. Existing career advisory solutions fail because they rely on static skills taxonomies that become obsolete upon release, or offer generic job boards that match candidates only to existing vacancies without evaluating future viability.

Crucially, individuals suffer from "reskilling paralysis" they lack visibility into which specific competencies are emerging versus declining, how different paces of AI disruption will alter role demand, and which accessible training avenues exist. Without explainable, scenario-tested guidance, workers risk investing time into obsolete training. Addressing this mismatch is essential for safeguarding national economic productivity (SDG 8) and ensuring equitable, lifelong access to relevant vocational skills (SDG 4).

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Aiversecity Future Skills is an explainable workforce navigation system that takes users through a transparent journey: CV/profile → future job trends → explainable role matches → skill gaps and verified learning pathways.

The solution operates through seven integrated modules:

1. Profile & CV Scan: Parses text-based PDFs or raw CV text into skills, experience, and industries. It highlights supporting CV excerpts, allowing users to review, edit, and validate detected skills before saving their profile.
2. Future Skills Radar: Tracks rising and declining competencies across tech and non-tech industries under three disruption scenarios Baseline, Moderate, and Fast to account for labor-market uncertainty.
3. Role Matches: Ranks suitable roles using transparent vector similarity, providing an explainable breakdown of matched competencies versus identified skill gaps.
4. Role Details: Details granular requirements, task automation risks, and industry context for any selected role.
5. Reskilling Pathway: Maps gaps to ordered learning plans prioritizing verified Malaysian resources. Adhering to strict AI guardrails, if an accredited course is unverified, the gap is transparently marked "uncovered" rather than hallucinating an invented link.
6. SDG Links: Explicitly connects every role and skill to UN Sustainable Development Goals.
7. Sources & About: Discloses underlying data sources, scenario assumptions, and AI guardrails.

Technical Implementation & Sponsor Utilization: The system uses rule-assisted parsing and embedding models for deterministic matching. The web application is built and hosted on Replit, while batch vector computations and disruption scenario indexing run on RunPod cloud instances. The platform keeps humans firmly in charge of career decisions while eliminating generative hallucinations.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend & Client Dashboard: Streamlit / Next.js deployed on Replit.
Backend Processing API: Python FastAPI orchestrating CV parsing, scenario weighting, and matching logic.
Database & Vector Store: SQLite / PostgreSQL with ChromaDB for storing role profiles, skills taxonomies, and verified course embeddings.
- AI & NLP Pipeline:
  -Extraction: Hybrid rule-based regex and open LLM / Gemini API with structured JSON output for excerpt-backed CV entity detection.
  -Matching Math: Cosine similarity over normalized skill embeddings.
- Sponsor Platform Utilization:
  -Replit: Primary hosting environment for rapid full-stack deployment, continuous integration, and live judge demonstrations.
  -RunPod: Cloud compute environment for hosting vector embedding generation and periodic trend-signal data aggregation pipelines.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Applying the Design Thinking framework, our team engaged with Malaysian jobseekers and TVET graduates experiencing automation anxiety.

Empathize & Define: Users expressed deep skepticism toward "black-box" AI career advice and frustration over hallucinated or expensive course links.
Ideate & Prototype: We rejected generic conversational chatbots in favor of an evidence-backed workflow. We implemented human-in-the-loop CV review (allowing users to confirm detected skills against source text) and introduced tri-scenario disruption modeling (Baseline, Moderate, Fast) to make labor uncertainty manageable.
Deployment Strategy for Phase 2:
Grounded Registry Expansion: Partner with Malaysian vocational institutions to expand verified HRD Corp-claimable and TVET micro-credentials. Unverified courses will remain strictly flagged as uncovered.
Scalable Cloud Infrastructure: Containerize backend ingestion routines on RunPod and serve the client on Replit for reliable, low-bandwidth nationwide accessibility.
Inclusivity & Local Relevance: Implement bilingual support (English and Bahasa Malaysia) to ensure TVET students and regional workers can navigate transitions smoothly.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: Target 4.4 ("By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship"). Secondary: Target 8.2 ("Achieve higher levels of economic productivity through diversification, technological upgrading and innovation").
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In Malaysia, while 56% of professionals actively use AI tools at work, only 26% have received formal training (Hays Asia Salary Guide 2025). Furthermore, 600,000+ workers face structural displacement over the next 3–5 years without accessible, verified reskilling pathways (MOSTI, 2024).

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	1. Verified Skills-Gap Identification Rate (%): Percentage of user profiles mapped to validated competencies with zero ungrounded skill assertions. 2. Course Provenance Ratio (%): Percentage of recommended courses mapped to verified, low-cost/HRD Corp registries (with unverified skills strictly marked as "uncovered" rather than hallucinated).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	1. 40% reduction in reskilling planning time compared to manual career counseling. 2. 100% citation transparency. 3. 75%+ user capability alignment toward resilient, high-growth SDG 4 and SDG 8 career paths in pilot tests.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

[/] Yes [ ] No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python, Streamlit, FastAPI, Pydantic, Pandas, NumPy, Scikit-learn, Plotly, PyPDF	Code (Open-Source Frameworks & Libraries)	PSF / Apache-2.0 / MIT / BSD-3-Clause
ChromaDB & SQLite / PostgreSQL	Code (Vector Store & Database Engine)	Apache-2.0 / Public Domain / PostgreSQL License
Google Gemini 1.5 Flash (gemini-1.5-flash)	Model (Pre-trained Generative AI API)	Google Generative AI API Terms
Sentence-Transformers (all-MiniLM-L6-v2) / text-embedding-004	Model (Pre-trained Embedding Model)	Apache-2.0 / Google API Terms
Aiversecity Synthetic CV & Medallion Pipeline Datasets (synthetic_cvs.csv, cv_experiences.csv, role_match_explanations.csv)	Dataset (Synthetic Project Dataset	MIT License (Internal Project Artifact)
WEF Future of Jobs, Hays Asia 2025, MOSTI Projections, HRD Corp Registry & UN SDG Metadata	Dataset (Public Benchmarks & Curated Course Registry)	Public Domain / Attribution

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

25 October 2026